The Journal of Chemical Physics

RESEARCH ARTICLE | NOVEMBER 15 1973

Accurate dissociation energies from rotational predissociation and long-range forces: $B\ ^1\Pi$ LiH

Kermit R. Way; William C. Stwalley

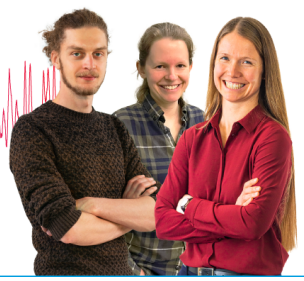
Check for updates

J. Chem. Phys. 59, 5298–5303 (1973)
https://doi.org/10.1063/1.1679873

View Online
Export Citation

01 August 2025 13:45:20

Webinar From Noise to Knowledge
May 13th – Register now
Zurich Instruments
Universität Konstanz



AIP Publishing

Accurate dissociation energies from rotational predissociation and long-range forces: $B\ ^1\Pi$ LiH*

Kermit R. Way† and William C. Stwalley‡
Department of Chemistry, University of Iowa, Iowa City, Iowa 52242
(Received 11 June 1973)

A new approach for determination of accurate dissociation energies from experimental predissociation data and the calculated long-range potential is described. In this case long-range predissociation in the rotational manifold of only a single vibrational level is required to yield a precise dissociation limit. The case of $B\ ^1\Pi$ LiH is considered in detail and a dissociation limit of 34 492.5 $\pm$ 0.5 cm^{-1} is obtained, 2.5 cm^{-1} lower and 4 times more precise than the best previous result. Also presented is a reasonably good representation of the potential energy curve for the $B\ ^1\Pi$ LiH molecule, constructed by an isotopically combined Rydberg–Klein–Rees method based on the concept of mass-reduced quantum numbers: $\eta \equiv [v+\frac{1}{2}]/\sqrt{\mu}$ and $\xi \equiv J(J+1)/\mu$.

I. INTRODUCTION

While predissociation has long been used in the determination of dissociation energies,[1,2] it is only recently that the close relationship between predissociation, the dissociation limit, and the long-range behavior of the internuclear potential has been used[3–10] (albeit somewhat uncritically[11–14]). Perhaps this is a result of the considerable progress made in the theory of long-range intermolecular forces in the past decade, evidenced by a number of recent reviews.[5,15–20]

Bernstein[3] pointed out that when the long-range behavior of the internuclear potential can be approximated by

$$V(R) = -C_n R^{-n} (C_n > 0, n > 2)\ , \tag{1}$$

the effective potentials (atomic units, $e = \hbar = m_e = 1$, are used throughout, except for some energies in cm^{-1})

$$U_J(R) \equiv V(R) + \frac{J(J+1)}{2\mu R^2} \tag{2}$$

will have maxima

$$U_J = \left(\frac{n-2}{2}\right) C_n^{2/(n-2)} \left(\frac{J(J+1)}{n\mu}\right)^{n/(n-2)} \tag{3}$$

at internuclear distances

$$R_J^\ddagger = \left[\frac{n\mu C_n}{J(J+1)}\right]^{1/(n-2)}\ , \tag{4}$$

the double dagger being adopted here to suggest the close analogy between these "centrifugal barriers" and the barriers of chemical kinetics.

It should also be noted that μ is the reduced mass of the nuclei (not atoms!) and that $\{J(J+1)\}$ should be replaced in general by the integer nearest $\{J(J+1) - 2\Omega^2 + \langle J_e^2 \rangle\}$, where J is then the total angular momentum; Ω, the component of J on the diatomic axis; and J_e, the sum of the angular momenta of all the electrons.[21]

Classically, any state with angular momentum J and positive energy $E < U_J^\ddagger$, and placed at a classically allowed internuclear distance $< R_J^\ddagger$, would be stable. Quantum mechanically, however, tunneling (unimolecular decomposition) can occur, so such states have a finite lifetime and energy width Γ and are in a strict sense part of the continuum. However, unless these so-called "quasibound" states are very near the top of the centrifugal barrier, they will behave primarily like bound states, with, for example, spectroscopic transitions into these states being no broader than those into adjacent bound states. Qualitatively, a quasibound state infinitesmally below $U_J^\ddagger$ would be expected to have about a 50% chance of decomposing every time it reached its outer turning point, so its lifetime τ should be of the same order as its vibrational period and its energy width Γ the order of $\hbar/\tau$. For a reasonable τ of 10^{-13} sec, $\Gamma \sim 50$ cm^{-1}. Such broad states would be difficult to observe spectroscopically (except perhaps by the resonance fluorescence technique), but should be observable as "orbiting resonances" in atom-atom scattering experiments.[3,22,23] What is usually observed spectroscopically is a series of rotational levels for vibrational level v which suddenly break off at some critical J_c (a monotonically decreasing function of v). One then assumes that

$$\epsilon(v, J_c) < U_{J_c}^\ddagger\ , \tag{5a}$$

$$\epsilon(v, J_c+1) > U_{(J_c+1)}^\ddagger\ , \tag{5b}$$

where the energy $\epsilon(v, J)$ is referred to the dissociation limit, and where $\epsilon(v, J_c+1)$ is not known experimentally, but only estimated by extrapolation. Since the vibrational–rotational levels are usually known with respect to the $v=0$, $J=0$ (or more generally $J=\Omega$) level, one obtains bounds on the dissociation energy $D_0 (\equiv -\epsilon(0, \Omega))$

$$E(v, J_c) - U_{J_c}^\ddagger \equiv \{\epsilon(v, J_c) - \epsilon(0, \Omega)\} - U_{J_c}^\ddagger < D_0\ , \tag{6a}$$

01 August 2025 13:45:20

$$E(v, J_c+1) - U^{\ddagger}_{(J_c+1)} \equiv \{\epsilon(v, J_c+1) - \epsilon(0, \Omega)\} - U^{\ddagger}_{(J_c+1)} > D_0 \ . \quad (6b)$$

The traditional approach[1,2] to predissociations governed by the centrifugal barrier (cases Ib and III of Herzberg, Ref. 2) has involved the limiting curve of dissociation (LCD). One plots the rotational levels $E(v,J_c)$ and $E(v,J_c+1)$ for several vibrational levels versus $J(J+1)$, and graphically takes the limit $J \to 0$ since $\lim_{J\to 0} U^{\ddagger}_J = 0$ (unless the potential has a barrier!). While this procedure may be more accurate than Birge–Sponer extrapolations, it has two disadvantages: the predissociation in the rotational manifolds of at least *three* different vibrational levels is required to insure that there is no barrier in the potential; and for light diatomics the LCD is known to fall significantly below the locus of barrier maxima (LBM).[23]

The approaches invoking long-range forces[3–10] have all (except Ref. 10) *assumed* $V(R) = -C_n/R^n$ [Eq. (1)] (and hence no barrier in the potential) but then predissociation in the rotational manifolds of only *two* vibrational levels is needed to determine the dissociation energy, the other determined parameter being C_n. However, the experimental application of all of this work to data may be criticized[11–14] since in no case (except possibly Se_2 and S_2; see Ref. 14) are the relevant centrifugal barriers at sufficiently large internuclear distances for Eq. (1) to hold accurately.

In this paper, we take a different point of view: we assume not merely the form, but also the magnitude of the long-range forces, in which case observation of predissociation in the rotational manifold of a *single* vibrational level is sufficient to precisely determine dissociation.

Thus as the theoretical information about the long-range potential increases, the amount of experimental data required reduces to the minimum: one observed vibrational band showing predissociation. A search of the literature has produced a case[24] with a single vibrational level showing rotational predissociation with the relevant centrifugal barriers clearly in the long range region: $v=2$ of $B^1\Pi$ LiH (relevant $R^{\ddagger}_J \geq 18a_0$). Note, however, that even here the R^{-8} term must be considered and Eq. (1) is invalid. In the present work, the long-range potential calculated from the properties of separated atoms is used to predict the appearance of the spectral lines. From this, extremely tight bounds on the dissociation energy are readily determined. Additional examples possibly amenable to such analysis are currently under study: $X^2\Sigma^+$ HgH ($v=4$),[25] $X^2\Sigma^+$ HgD ($v=6$),[26] and $X^1\Sigma_g^+$ D_2 ($v=21$).[27]

II. THE POTENTIAL

The potential energy curve for $B^1\Pi$ LiH (Fig. 1) was obtained by combining an RKR[28] potential well with the long range tail and an exponential inner wall.

The RKR potential was constructed using data for the two isotopic variants ^{7}LiH and ^{7}LiD taken from the work of Velasco.[24] The data were combined using the concept[29,30] of mass-reduced quantum numbers for vibration

$$\eta \equiv (v+\tfrac{1}{2})/\sqrt{\mu} \simeq (\pi\sqrt{2})^{-1} \int [E - V - \xi/(2R^2)]^{1/2} dR \quad (7a)$$

and for rotation

$$\xi \equiv J(J+1)/\mu \ , \quad (7b)$$

which will be discussed in detail elsewhere.[30] Suffice it to say that since different isotopic variants of a given diatomic have to a very good approximation the same potential curve and since the semiclassical quantization condition [given in Eq. (7a)] is accurately obeyed even for the lightest molecules, virtually all equations [e.g., Eqs. (2), (3), (4), and (7a) of this paper, the Dunham expansion,

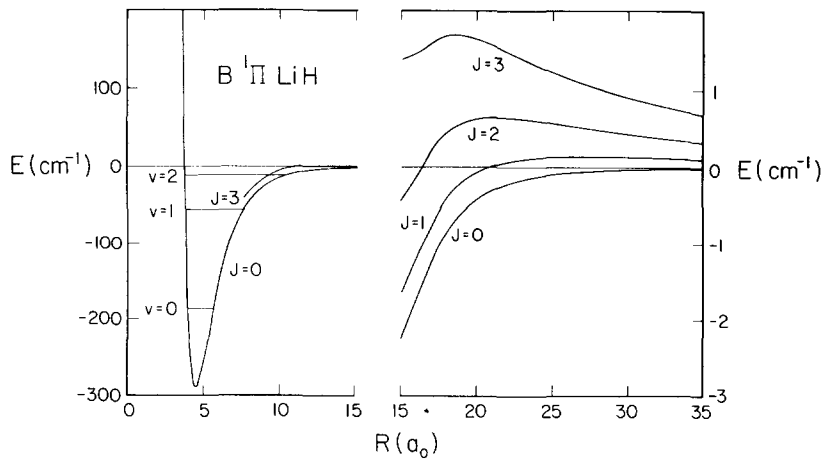


FIG. 1. The potential G (as described in text) and corresponding effective potentials for the $B\,^1\Pi$ state of LiH.

01 August 2025 13:45:20

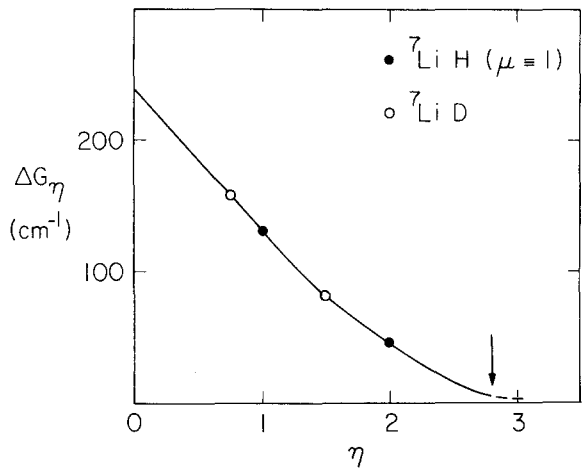


FIG. 2. The mass-reduced vibrational spacing $\Delta G(\eta)$ data (● for ^{7}LiH, ○ for ^{7}LiD) and fit (solid curve) for the $B^1\Pi$ state using $\mu(^7\text{LiH}) \equiv 1$.

the RKR equations] and plots (e.g., LCD, LBM, Birge–Sponer) involving states or their expectation values in a single potential curve can be recast in mass-reduced form which combines data for all isotopic variants of a given diatomic. In the present case of $B^1\Pi$ LiH this combining of isotopic data is essential for obtaining a meaningful curve.

More explicitly, the mass-reduced vibrational spacing function ("Birge-Sponer plot", Fig. 2)

$$\Delta G(\eta) = \sqrt{\mu}\,\Delta G(v+\tfrac{1}{2}) \cong \partial E(\eta, \xi)/\partial \eta \,\big|_{\xi=0} \qquad (8)$$

and the mass-reduced rotational constant function (Fig. 3)

$$B(\eta) = \mu B(v) \equiv \partial E(\eta, \xi)/\partial \xi \,\big|_{\eta} \qquad (9)$$

are better defined than ΔG and B for any individual isotopic combination, and thus allow a more precise determination of the potential.[29,30] In this case, the number of vibrational spacings and rotational constants are doubled and the resulting cubic spline[31] fits of Figs. 2 and 3 are much better determined. The RKR potential was then calculated in steps of $\Delta\eta = 0.1$ [$\mu(^7\text{LiH}) \equiv 1$] up to $\eta = 2.8$, and the resultant values of the turning points also fit to a cubic spline.[31] The inner wall near and above dissociation was obtained by extrapolation of the RKR inner turning points using an exponential function.

The long-range portion of the $B^1\Pi$ LiH potential was found to be

$$V(R) = D_0 - 84(\pm 10)/R^6 - 14\,800(\pm 1000)/R^8 \qquad (10)$$

using an extension[32] of the work of Mahan.[33] This method (originally applied to ground and excited alkali-rare gas C_6 values only) uses the fact that the energies of the transitions whose oscillator strengths dominate the alkali metal polarizabilities are below those same energies in many other atoms. In particular, they are lower than the corresponding energies in the H atom as well as the rare gases. Note that LeRoy's criterion[14] for the distance at which exchange becomes significant [and Eq. (10) invalid] yields $R = 14a_0$ for $B^1\Pi$ LiH, well inside the centrifugal barriers under consideration; and that the R^{-8} term (which lowers $U_J^\dagger$ and increases $R_J^\dagger$) is a significant correction. Using the long-range potential of Equation 10, one obtains centrifugal barriers $U_2^\dagger = 0.63(\pm 0.03)$ cm^{-1} at $R_2^\dagger = 21.3 \times (\pm 0.5)a_0$; and $U_3^\dagger = 1.70(\pm 0.07)$ cm^{-1} at $R_3^\dagger = 18.4 \times (\pm 0.4)a_0$ (the corresponding numbers from Eqs. (3) and (4) using only the R^{-6} term are 0.75 cm^{-1}, $19.1a_0$; 2.11 cm^{-1}, $16.1a_0$). The long-range potential was joined to the RKR potential by a cubic spline[31] at $R = 17a_0$.

It is also worth noting that only the data corresponding to the $B^1\Pi^-$ component were used in computing the RKR curve since the $B^1\Pi^+$ component should be slightly perturbed by the $A^1\Sigma^+$ state, especially at large distances.[34,35]

Finally, although we have here gone to some effort to obtain an accurate total potential, much of the analysis of the following sections is insensitive to most of it, depending strongly (exponentially) only on the region near the relevant centrifugal barriers. A few copies of the Fortran subroutine which generates this potential (labeled G below) are available from the authors. In addition to this benchmark potential, some slightly modified potentials (designated $G \pm n$) were constructed by raising/lowering the asymptote D_0 with respect to the RKR

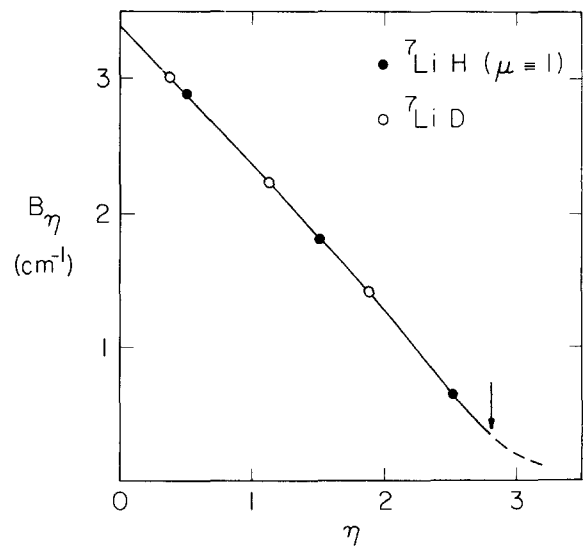


FIG. 3. The mass-reduced rotational constant $B(\eta)$ data (● for ^{7}LiH, ○ for ^{7}LiD) and fit (solid curve) for the $B^1\Pi$ state using $\mu(^7\text{LiH}) \equiv 1$.

01 August 2025 13:45:20

well by n cm^{-1} and rejoining at $17a_0$. The effect of this kind of modification is to shift the predissociating energy levels down or up with respect to the barrier maxima, while keeping the shape and height of the centrifugal barriers essentially fixed, with linewidth depending strongly on this shift.

It should be noted that the energy level shift is *not* by $\mp n$ cm^{-1} for the $G \pm n$ potential.

III. CALCULATION OF PREDICTED PREDISSOCIATION SPECTRA

One estimate[23,36] of the spectroscopic linewidth of a quasibound level is

$$\Gamma = \frac{2}{[\partial\eta/\partial E]_{\max}}, \tag{11}$$

where η is the scattering phase shift which can be calculated using, for example, the method of Gordon.[37] The energy for the quasibound level is taken as the point where $\partial\eta/\partial E$ is a maximum. Energies of the bound levels are found using an eigenvalue program[38] employing the predictor–corrector method. Linewidths predicted by this method for the potential, G, and the modifications of it are shown in Table I.

While this estimate is sufficient for sharp levels, for broader levels (particularly close to the dissociation limit where $\epsilon \approx \Gamma$), the actual shapes of the lines expected for a particular potential should be obtained using a bound-to-free intensity program.[38] Here the intensity is calculated for a Q branch ($\Delta J = 0$) transition from $v = 2$, $J = 2$ or 3 of the ground state[39] to a series of energies just above the asymptote of the B state. By plotting these calculated intensities, one produces the spectra in Figure 4, where the scale differences should be noted.

IV. COMPARISON WITH REPORTED SPECTRA

Velasco reports the $v = 2$, $J = 2$ level is sharp while the $J = 3$ level is not seen.[24] However, there is no indication either of his criteria for sharpness or how broad a feature he would have reported.

TABLE I. Calculated energies with respect to the asymptote $\epsilon(v, J)$ and also widths $\Gamma(v, J)$ (each in cm^{-1}) of the $v = 2$, $J = 2$ and 3 bound or quasibound levels, for various trial potentials (see text for notation) with dissociation energies $D_0 \equiv -\epsilon(0,1)$, representing the $B\,^1\Pi$ state of LiH.

Potential	$\epsilon(2,2)$	$\Gamma(2,2)$	$\epsilon(2,3)$	$\Gamma(2,3)$	$D_0 \equiv -\epsilon(0,1)$
$G-3$	0.59	0.47	a	a	170.35
$G-2$	0.46	0.23	a	a	171.35
$G-1$	0.29	0.076	2.2	2.9	172.35
G	0.14	0.013	2.19	2.5	173.35
$G+1$	−0.31	bound	2.02	1.66	174.35
$G+2$	−0.92	bound	1.74	0.94	175.35
$G+4$	−2.44	bound	0.86	0.11	177.35

[a]Not calculated.

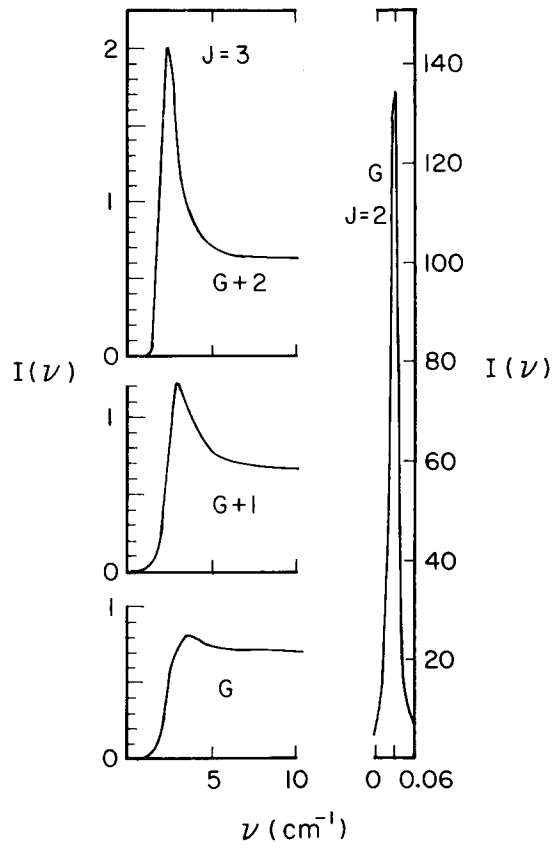


FIG. 4. The calculated line shapes for the Q branch of the $X \rightarrow B$ transition of ^{7}LiH for the potential G and two modifications. Note the different scales used.

LeRoy and Bernstein[23] indicate the sharp vs broadened distinction should be in the range

$$0.05 \text{ cm}^{-1} < \Gamma < 0.25 \text{ cm}^{-1} . \tag{12}$$

At the other extreme, from examination of rotational predissociation spectra, it appears that levels as wide as 3 cm^{-1} can be seen.

Thus Table I indicates that broadening would almost certainly have been noticed if the $J = 2$ level were more than 0.50 cm^{-1} above the asymptote. This establishes a *lower* bound on the asymptote. However, such a calculation of peak sharpness is not a sufficient criterion to explain the absence of the $J = 3$ line in the reported spectrum.

In Fig. 4, the areas of all of the peaks (including $J = 2$, G) are equal within a factor of ~ 2 when integrated down to the baseline and over a width of 2 cm^{-1}. Since the $J = 2$ transitions are observed, the $J = 3$ transitions are also intense enough to be seen. If one knew that had a feature such as the $J = 3$ G curve of Fig. 4 been present, it would have been seen and reported, he would have an extremely tight upper bound on the dissociation energy. No

01 August 2025 13:45:20

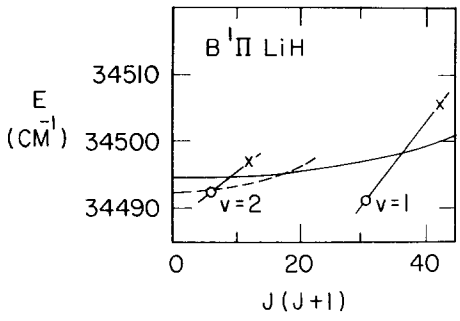


FIG. 5. The LCD curve (solid) and rotational predissociation data (O for last observed, × for first unobserved) of Velasco[24] compared to the asymptotic LBM curve (dashed) implied by the present work.

such feature was seen in the original work[24]; recently, Velasco re-examined his plates and confirmed this point.[40] Thus although there is reason to believe the G potential provides an upper bound to the asymptote, it seems much more certain that a peak of the shape shown as $G+1$ would have been observed. If the latter is true, the $J=2$ reference level cannot be bound by more than 0.31 cm^{-1}.

In summary then, the $J=2$ level at 34492.61_5 cm^{-1} above the ground state zero point level[24] cannot be more than 0.50 cm^{-1} above dissociation nor more than 0.31 cm^{-1} below dissociation. Allowing for the uncertainty in the spectrum itself and in the long range coefficients, one obtains a final value for the dissociation limit of 34492.5 ± 0.5 cm^{-1} over the (0, 0) level of the ground state which is significantly lower and more certain than Velasco's value[24] of 34495 ± 2 cm^{-1}. The corresponding D_0 values for ^{7}LiH are thus 19588.7 ± 0.5 cm^{-1} in the ground state, 8549.4 ± 0.5 cm^{-1} in the $A^1\Sigma^+$ state and 174.4 ± 0.5 cm^{-1} in the $B^1\Pi$ state. It might also be noted that 34492.5 cm^{-1} is the dissociation limit corresponding to the G potential.

V. CONCLUSIONS

If the potential curve of an electronic state of a molecule can be investigated spectroscopically to long range, the present method of treatment of data offers an appreciably more accurate value for the dissociation energy than does the simple visual estimate of the limiting curve of dissociation. This is illustrated in Fig. 5, where the LCD and energy levels given by Velasco[24] are compared with the asymptotic LBM (valid *only* for distances corresponding to $J(J+1) \lesssim 20$) indicated in this work. Note, however, that it does require as additional input the long-range potential which cannot always be accurately obtained, particularly in excited states. Further, it demonstrates the usefulness of including an indication of linewidths (as, for example, was recently done by DiLonardo and Douglas[41]) in published spectra. Finally, if used by spectroscopist, this method might enable him to find a feature similar to the $J=3$, G line shape in Fig. 4. Knowledge of whether or not such a line shape was present would very much reduce the (small) remaining uncertainty.

The present work also illustrates the utility of the concept of mass-reduced quantum numbers in the construction of potential curves.

*Acknowledgment is made to the Donors of the Petroleum Research Fund, administered by the American Chemical Society, for support of this work.

†NASA Trainee (1971–1972); Phillips Petroleum Research Fellow (1972–1973).

‡Alfred P. Sloan Foundation Fellow.

[1]A. G. Gaydon, *Dissociation Energies and Spectra of Diatomic Molecules* (Chapman and Hall, London, 1968), 3rd. ed.
[2]G. Herzberg, *Spectra of Diatomic Molecules* (Van Nostrand, 1950), 2nd ed.
[3]R. B. Bernstein, Phys. Rev. Lett. **16,** 385 (1966).
[4]M. A. Byrne, W. G. Richards and J. A. Horsley, Mol. Phys. **12,** 273 (1967).
[5]T. Y. Chang, Rev. Mod. Phys. **39,** 911 (1967).
[6]T. Y. Chang, Mol. Phys. **13,** 487 (1967).
[7]P. G. Burke, J. H. Tait and A. Dalgarno, Chem. Phys. Lett. **1,** 345 (1967).
[8]J. A. Horseley and W. G. Richards, J. Chim. Phys. **66,** 41 (1969).
[9]J. M. Ricks and R. M. Barrow, Can. J. Phys. **47,** 2433 (1969).
[10]O. Goscinski and O. Tapia, Mol. Phys. **24,** 641 (1972).
[11]W. C. Stwalley and H. Kramer, J. Chem. Phys. **49,** 5555 (1968).
[12]W. C. Stwalley, "A Critical Examination of the Relation of Long-Range Forces to the Interpretation of Diatomic Spectra near Dissociation", Paper D12 at the 25th Symposium on Molecular Structure and Spectroscopy, Columbus, OH, September, 1970 (unpublished).
[13]W. C. Stwalley, Chem. Phys. Lett. **7,** 600 (1970).
[14]R. J. LeRoy, "Energy Levels of a Diatomic Near Dissociation: Dependence on the Determination of the Long-Range Potential", to be published in a Specialist Periodical Report of the Chemical Society (London) on Electronic Spectroscopy. This review includes and develops essentially all points discussed in Ref. 12, as well as many others. One of us (W.C.S.) wishes to thank Professor LeRoy for including these points.
[15]A. Dalgarno and W. D. Davidson, Adv. At. Mol. Phys. **2,** 1 (1966).
[16]J. O. Hirschfelder and W. M. Meath, Adv. Chem. Phys. **12,** 3 (1967).
[17]A. D. Buckingham, Adv. Chem. Phys. **12,** 107 (1967).
[18]A. Dalgarno, Adv. Chem. Phys. **12,** 143 (1967).
[19]H. Margenau and N. R. Kestner, *Theory of Intermolecular Forces* (Pergamon, London, 1971), 2nd. ed.
[20]P. R. Certain and L. W. Bruch, in *MTP International Review of Science, Physical Chemistry Section, Theoretical Chemistry Volume* (MTP Medical and Technical Publishing Co., Oxford, England, to be published); also available as University of Wisconsin Theoretical Chemistry Institute Report WIS-TCI 460 (1971).
[21]R. T Pack and J. O. Hirschfelder, J. Chem. Phys. **49,** 4009 (1968); J. Chem. Phys. **52,** 521 (1970); J. Chem. Phys. **52,** 4198 (1970).
[22]W. C. Stwalley, *Proceedings of the Sixth International Conference on the Physics of Electronic and Atomic Collisions* (MIT Press, Cambridge, MA, 1969), p. 51.

01 August 2025 13:45:20

[23]R. J. LeRoy and R. B. Bernstein, J. Chem. Phys. **54,** 5114 (1971).
[24]R. Velasco, Can. J. Phys. **35,** 1204 (1957).
[25]D. M. Eakin and S. P. Davis, J. Mol. Spectrosc. **35,** 27 (1970).
[26]W. C. Stwalley, unpublished analysis of spectra taken by Dr. D. M. Eakin and Professor S. P. Davis.
[27]H. Bredohl and G. Herzberg, Can. J. Phys. **51,** 867 (1973).
[28]See, for example, R. N. Zare, J. Chem. Phys. **40,** 1934 (1964). The authors wish to thank Professor Zare for making a recent version of this computer program available to them.
[29]Here we ignore the various small contributions to the potential of the order of m_e/μ which make the potential for LiH very slightly different than that for LiD; see Ref. 30.
[30]W. C. Stwalley, "The Utility of Mass-Reduced Quantum Numbers", paper AA5 at the 27th Symposium on Molecular Structure and Spectroscopy, Columbus, OH, June, 1972 (unpublished).
[31]We wish to thank Professor K. Kaiser for making his "tensioned spline" program available to us.
[32]W. C. Stwalley, work in progress.
[33]G. D. Mahan, J. Chem. Phys. **48,** 950 (1968); J. Chem. Phys. **50,** 2755 (1969).
[34]See, e.g., the analogous case of OH in Ref. 14. Helpful discussions and correspondence on this point with Professor R. T Pack are gratefully acknowledged.
[35]Even though this perturbation seems relatively weak [i.e., the Λ doubling ($\lesssim 0.2$ cm^{-1}) in $B\ ^1\Pi$ is barely observable], the consequences could be particularly significant in $v = 3$ of LiD (thus far unobserved) since this very weakly bound level could have rotational predissociation occuring at different J_c in $B\ ^1\Pi^+$ than in $B\ ^1\Pi^-$.
[36]M. S. Child, "Diatomic Predissociation Line Widths", to be published in a Specialist Periodical Report of the Chemical Society (London) on Electronic Spectroscopy; also available as University of Wisconsin Theoretical Chemistry Institute Report WIS-TCI-416 (1973).
[37]R. G. Gordon, J. Chem. Phys. **51,** 14 (1969). The program is available as No. 187 through the Quantum Chemistry Program Exchange, Indiana University.
[38]The authors wish to thank Professor K. Sando for making available his eigenvalue and bound-to-free intensity program. See K. Sando, Mol. Phys. **23,** 413 (1972).
[39]The ground state potential is from R. J. Fallon, J. T. Vanderslice, and E. A. Mason, J. Chem. Phys. **32,** 1453 (1960).
[40]R. Velasco (private communication).
[41]G. DiLonardo and A. E. Douglas, Can. J. Phys. **51,** 434 (1973).

01 August 2025 13:45:20